DevOps aims to accelerate the delivery of high-quality products and services by unifying software development (Dev) and software operations (Ops).

Top reasons behind the global adoption of DevOps

- **Accelerates and extends the Agile methodology:** Many global organisations adopt the Agile methodology to accelerate software delivery and improve software quality. This unifies software development and testing activities. With DevOps, the Agile methodology can be extended by integrating software development, testing and development activities, thereby, speeding up software delivery by removing the bottlenecks.
- **Strong communication and collaboration:** With DevOps, programmers, testers and operations test engineers work as a single team. In this environment, every member has a strong understanding of pre-defined business requirements and organisational goals, which they can attain by working in a collaborative manner, focusing on the common goal rather than on individual goals. It creates a strong environment for experimentation, innovation and research.
- **Automation of repetitive tasks:** With DevOps, organisations adopt automation tools and strong production platforms that help operations engineers to deploy the software applications without investing extra time and effort. The testing professionals can use test automation tools to evaluate the software under varied user conditions and produce more accurate test results. The automation of common and routine tasks helps businesses to reduce development cycles and shorten delivery cycles.
- **Reduction in failures:** DevOps is focused on shorter development cycles and frequent code releases. As the programmers focus on a specific feature or component of the software application, they can easily write high-quality code and repair common coding errors. At the same time, the testers can detect the flaws in the software and get them repaired without any delay. Hence, enterprises can reduce failures and fixes significantly by adopting DevOps.

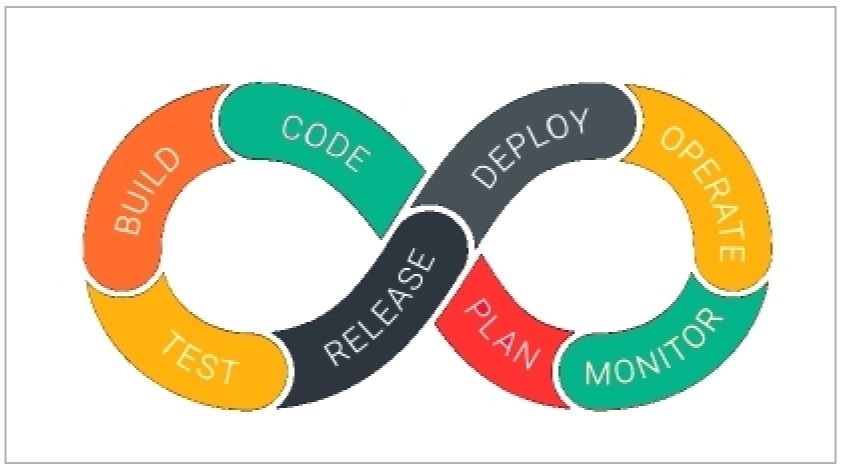

Figure 1: The DevOps life cycle

- **Creating a performance-based work environment:** DevOps enables enterprises to modernise and transform the work environment by focusing only on performance. The unified work environment makes various stakeholders in the project collaborate with each other without bureaucratic obstacles. Also, software developers and operations engineers can perform optimally by understanding their roles and sharing risks.

DevOps life cycle

The DevOps life cycle comprises various stages and is highlighted in Figure 1. The terms in this figure are explained below.

- **Plan:** Initial planning with regard to the type of application required
- **Code:** Coding the application as per the client's requirements with proper planning
- **Build:** Building the application by integrating the various types of code created by diverse programmers in the coding phase
- **Test:** Testing of the application built so far to verify and validate the requirements, and re-build the applications if components are not working effectively
- **Releases:** Deploying the application in live scenarios
- **Deploy:** Deployment of the cloud in cloud environments for additional usage
- **Operate:** Performing operations on code, if any
- **Monitor:** Strong monitoring of the application as per the client's requirements and validating the end user experience

The roles and responsibilities of team members in DevOps

DevOps engineer: This person needs to have a core understanding of the SDLC and in-depth knowledge of various automation tools for developing digital pipelines (CI/CD).

Roles and responsibilities

- Apply cloud (AWS, Azure, GCP) computing skills to deploy upgrades and fixes
- Design, develop and implement software integrations based on user feedback
- Troubleshoot production issues and coordinate with the development team to streamline code deployment
- Implement automation tools and frameworks (CI/CD pipelines)
- Analyse code and communicate detailed reviews to development teams to ensure a marked improvement in applications and the timely completion of projects
- Collaborate with team members to improve the company's engineering tools, systems, procedures and data security
- Optimise the company's computing architecture